

ELEX: Supporting hundreds of thousands of concurrent chat users

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ABOUT ELEX

With Google Cloud Engine and Google Cloud Translation API, ELEX is a global chat service key to the success of mobile massively multiplayer online game *Clash of Kings*.



About ELEX

Founded in 2008, ELEX publishes and generates revenue from popular mobile games. The business has more than 500 employees in its Beijing headquarters and in offices in Hefei, Shanghai, Hong Kong, Taiwan, and São Paulo in Brazil.

Industries: Media & Entertainment

Location: China

Google Cloud Results

- Reduced infrastructure costs for the *Clash of Kings* online chat service by 20%
- Helped create a community of players that drives the success of the massively multiplayer online strategy game
- Gained the scalability to seamlessly support peaks and troughs in demand

Real-time
for 30
concurrent
users per
second

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Translation API

(<https://cloud.google.com/translate/>)

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(<http://en.elex-tech.com/Index/about>) publishes and generates revenue from popular mobile games. The

business has more than 500 employees in its Beijing headquarters and in offices in Hefei, Shanghai, Hong Kong, Taiwan, and São Paulo in Brazil. ELEX has grown rapidly and enjoyed considerable success with annual turnover around US\$700 million in 2016.

One of ELEX's best-known products is the mobile massively multiplayer online strategy game *Clash of Kings* (<http://f.elex.com/forums/1-Clash-of-Kings>). The game features virtual battles, contests for thrones, and fights with dragons as players compete to build empires and become kings.

Published by ELEX in 2014, *Clash of Kings* is available on the Android and iOS mobile platforms. The game has been downloaded about 200 million times and is one of the top-grossing apps on Google Play and the App Store. ELEX's revenue comes from in-app purchases that can help players boost their standing in the game.

ELEX attributes the success of the game to "launching the right product at the right time with the right marketing strategy," says Strategic Partner and Business Development Director, ELEX. "We plan to continue developing *Clash of Kings* and run the game as a long-term game."

ELEX's technology strategy is based on identifying opportunities and problems, and then determining the role technology can play in the response. For example, *Clash of Kings* is played by people around the world who may speak different languages, but who need to chat with each other to plan and execute strategies to dominate their rivals.

“Google Cloud Platform had the worldwide presence we needed and, as a cloud platform, required minimal administration by our internal team. It could also meet our scalability and peak usage requirements, while being reliable enough to support a critical element of *Clash of Kings*.”

—Hu Chao, Product Manager, ELEX

“For example, we need to enable a player in England to communicate with a player in China who speaks

Mandarin Chinese,” says Hu Chao, Product Manager, ELEX. “So, we need a system that can translate English to Mandarin Chinese and Mandarin Chinese to English in real time.”

The translation tool initially selected by ELEX performed well initially but was not stable or scalable enough to support *Clash of Kings*’ fast-growing user community from around the world. The business began looking around for a more resilient, reliable tool.

The rapid growth and scale of *Clash of Kings*’ user community was also straining the servers, networks, storage, and associated systems used to run the game’s chat service. ELEX needed the platform used to run the chat service to be able to scale to support up to 300,000 concurrent users per second without compromising performance or reliability. The platform also had to scale down during lower demand to help ensure ELEX was not paying for capacity that remained unused in off-peak periods.

Furthermore, ELEX wanted to reduce the number of team members that had to dedicate time to system administration, so they could focus instead on game development. Finally, ELEX needed a service that had a presence all over the world to remain close to communities of *Clash of Kings* players in various countries and regions.

The selection of a new translation service by ELEX gained momentum as players began asking the company to enable them to cut text from chat messages and paste it into [Google Translate](#)

(<https://translate.google.com/>). The gamers believed this would enable them to obtain a better translation than the incumbent translation tool provided.

To validate the performance and functionality of Google's translation tools, ELEX selected 100 messages from its users and used a range of tools, including [Google Cloud Translation API](https://cloud.google.com/translate/) (<https://cloud.google.com/translate/>), to translate them.

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High quality real-time translations

“We found Google Cloud Translation API provided the best real-time translation results and Google the most reliable service,” says Chao. “The translation tools also integrated seamlessly with the *Clash of Kings* chat service.”

The business also selected Google Compute Engine (<https://cloud.google.com/compute/>) to run the chat module. “Google Cloud Platform (<https://cloud.google.com/>) had the worldwide presence we needed and, as a cloud platform, required minimal administration by our internal team,” says Chao. “It could also meet our scalability and peak usage requirements, while being reliable enough to support a critical element of *Clash of Kings*.”

Providing real-time chat to 300,000-plus users per second

Migrating to Google Compute Engine and Google Cloud Translation API has enabled ELEX to continue to provide real-time chat to more than 300,000 concurrent users per second, and allowed gamers from different countries, who use different languages, to compete with each other.

By moving to Google Compute Engine, ELEX has been able to move from a laborious process that

required the operations team to purchase and configure servers, to creating new virtual machine instances on demand. The business has also been able to add and close down these instances in line with traffic loads. “For example, when we have a lot of users online and sending messages, Google Cloud Platform will enable us to create new instances to support and balance the load,” says Chao.

Deploying the chat service on Google Compute Engine has also allowed ELEX to focus its team on delivering efficiencies and adding value to the business by creating new features for its games. This will, in turn, attract more consumers and generate more revenue for the business. Thanks to the ability to deploy instances on demand, the business has reduced the infrastructure costs of running that chat service by about 20%.

“Google Compute Engine has enabled us to make the best use of our human capital and reduce our server costs, while providing a strong, stable chat feature that enables players to communicate with each other,” says Chao. “In addition, Google Cloud Translation API has enabled us to serve our game to users all over the world without having to consider restrictions in certain markets due to communication problems. This is how Google has helped our company achieve its business goals.”